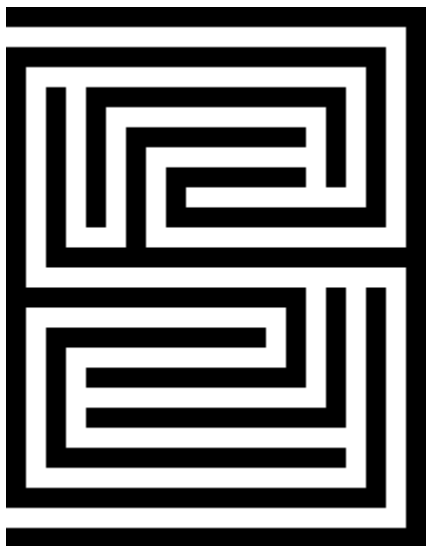




Wolves Finance Security Review Public Report

PROJECT: Wolves Finance Security Review

June-July 2021

Prepared For:

Wolves Finance Protocol

Wolves Finance

Prepared By:

Jonathan Haas | Bramah Systems, LLC.

jonathan@bramah.systems



Table of Contents

Executive Summary	3
Scope of Engagement	3
Timeline	3
Engagement Goals	3
Contract Specification	3
Overall Assessment	3
Timeliness of Content	5
General Recommendations	6
Long numbers can be written in scientific notation	6
Unused variables should be removed	6
Multiple TODO items remain	6
farmName capitalization can lead to registration concerns	7
Specific Recommendations	8
Private API should be removed	8
Deadline in swapExactETHForTokens could be declared constant	8
Setters should emit an event	8
Numerous typographical errors	9
Unbound for loops can lead to resource exhaustion	9
_getStableCoinInfo can be clarified	9
Commented out code should be removed	10
Toolset Warnings	11
Overview	11
Compilation Warnings	11
Test Coverage	11
Static Analysis Coverage	11
Directory Structure	12



Wolves Finance Protocol Review

Executive Summary

Scope of Engagement

Bramah Systems, LLC was engaged in May of 2021 to perform a comprehensive security review of the Wolves Finance smart contracts (specific contracts denoted within the appendix). Our review was conducted over a period of four business days by both members of the Bramah Systems, LLC. executive staff.

Bramah Systems completed the assessment using manual, static and dynamic analysis techniques.

Timeline

Review Commencement: May 13, 2021

Report Delivery: May 17th, 2021

Engagement Goals

The primary scope of the engagement was to evaluate and establish the overall security of the Wolves Finance protocol, with a specific focus on trading actions. In specific, the engagement sought to answer the following questions:

- Is it possible for an attacker to steal or freeze tokens?
- Does the Solidity code match the specification as provided?
- Is there a way to interfere with the contract mechanisms?
- Are the arithmetic calculations trustworthy?

Contract Specification

Specification was provided in the form of code comments. The contracts were provided via GitHub (commit hash [ee59f8d7964c3bf943a51f708a064835af2097eb](#))



Overall Assessment

Bramah Systems was engaged to evaluate and identify any potential security concerns within the codebase of the Wolves Finance protocol. During the course of our engagement, Bramah Systems found multiple instances wherein the team deviated materially from established best practices and procedures of secure software development within DLT. The team has since addressed these issues with a resolution or risk acceptance in a satisfactory manner and has taken steps to further their security resiliency.



Disclaimer

As of the date of publication, the information provided in this report reflects the presently held, commercially reasonable understanding of Bramah Systems, LLC.'s knowledge of security patterns as they relate to the Wolves Finance Protocol, with the understanding that distributed ledger technologies ("DLT") reWolvesn under frequent and continual development, and resultantly carry with them unknown technical risks and flaws. The scope of the review provided herein is limited solely to items denoted within "Scope of Engagement" and contained within "Directory Structure". The report does NOT cover, review, or opine upon security considerations unique to the Solidity compiler, tools used in the development of the protocol, or distributed ledger technologies themselves, or to any other matters not specifically covered in this report.

The contents of this report must NOT be construed as investment advice or advice of any other kind. This report does NOT have any bearing upon the potential economics of the Wolves Finance protocol or any other relevant product, service or asset of Wolves Finance or otherwise. This report is not and should not be relied upon by Wolves Finance or any reader of this report as any form of financial, tax, legal, regulatory, or other advice.

To the full extent permissible by applicable law, Bramah Systems, LLC. disclaims all warranties, express or implied. The information in this report is provided "as is" without warranty, representation, or guarantee of any kind, including the accuracy of the information provided. Bramah Systems, LLC. makes no warranties, representations, or guarantees about the Wolves Finance Protocol. Use of this report and/or any of the information provided herein is at the users sole risk, and Bramah Systems, LLC. hereby disclaims, and each user of this report hereby waives, releases, and holds Bramah Systems, LLC. harmless from, any and all liability, damage, expense, or harm (actual, threatened, or claimed) from such use.

Timeliness of Content

All content within this report is presented only as of the date published or indicated, to the commercially reasonable knowledge of Bramah Systems, LLC. as of such date, and may be superseded by subsequent events or for other reasons. The content contained within this report is subject to change without notice. Bramah Systems, LLC. does not guarantee or warrant the accuracy or timeliness of any of the content contained within this report, whether accessed through digital means or otherwise.

Bramah Systems, LLC. is not responsible for setting individual browser cache settings nor can it ensure any parties beyond those individuals directly listed within this report are receiving the most recent content as reasonably understood by Bramah Systems, LLC. as of the date this



report is provided to such individuals.

General Recommendations

Best Practices & Solidity Development Guidelines

Long numbers can be written in scientific notation

Multiple numbers throughout the protocol possessed a number of zeros. These numbers can be shortened for clarity's sake.

Resolution:

Numbers have been shortened in `a9793ed1fcd57dead72ad266d76d8045ac004a92`.

Unused variables should be removed

`Controller.sol` references an “**amount**” variable that should be removed.

Resolution:

The unused variable has been removed in `a56770dd3f70610378331f3e53d93a5c2308bf31`.

Multiple TODO items remain

A number of **TODO** items remain in the source code:

`./src/token/TradeFloor.sol:521:` * TODO: Provide the possibility to swap into WOWS

`./src/token/WOWSErc1155.sol:19:` * TODO's:

`./src/token/interfaces/IWOWSCryptofolio.sol:14:` * TODO: Describe cryptofolios



```
./src/token/interfaces/IWOWSERC1155.sol:14: * TODO: Describe cryptofolios
```

```
./src/investment/UniV2StakeFarm.sol:48: // TODO: Remove next 2 lines after dapp launch
```

(special reward condition)

```
./src/investment/UniV2StakeFarm.sol:187: // TODO: Remove after launch
```

```
./src/investment/UniV2StakeFarm.sol:212: // TODO: Remove after launch
```

Resolution:

One TODO was completed in **5c464bed44280b1b0442f1a988093bdfa88ab119**. The remaining TODOs were removed as being stale or irrelevant in **d720e0eb5fe5db54adaaebb61fff0e7cbe3da0df**.

farmName capitalization can lead to registration concerns

Controller.sol references an “**registerFarm**”, which allows registration of various farm information. However, the farm name logic (as displayed below) relies upon encoding the farm name, wherein different capitalized names will not produce the same bytes output, despite functionally being the same name.

```
// If we have one with same name, deactivate old one
```

```
bytes32 farmName = keccak256(abi.encodePacked(IFarm(_farmAddress).farmName()));
```

Resolution:

In consideration of the recommendation, and after weighing the design tradeoffs, case-sensitivity has been kept a design requirement and clearly documented in

431818f811ea35bf4d93cd3d172413a36787b2c8 to avoid future confusion.



Specific Recommendations

Unique to the Wolves Finance Protocol

Private API should be removed

There is a test role granted to access the private API for testing that should be removed.

bytes32 public constant TESTER_ROLE = 'test';

Resolution:

The private API has been removed in **0efbb0ac55dad8d5c6e08162c02ab68523594923**.

Deadline in swapExactETHForTokens could be declared constant

The 3600 deadline in the **swapExactETHForTokens** call in **swapIntoRewardToken** could be declared constant.

Resolution:

The deadline has been declared constant in **07e6c7756f210b1d2f29bdd2e9b44f0f3ce324f7**.

Setters should emit an event

There are numerous functions (such as **restrictTrading**) that limit users ability to interact with the contracts and should be made clear through event emittance.

Resolution: Events were added in **b22ba171eda426b993a82741ab15e70709d0afc5** and **099063860ccf4138dc549ffe3336f49a45fb7244**.



Numerous typographical errors

Numerous typographical errors exist throughout the protocol, namely constrained to code comments. We suggest the usage of a spell-check program be run upon the Solidity files to denote relevant issues.

Resolution:

Spelling and grammar corrections were made in `72aeb8e473ab910fe86f99f17171bfc830d7733c` and `cfd360fdbf4e74f316779b379ec5206815f69437`.

Unbound for loops can lead to resource exhaustion

Throughout the protocol, there are a number of for loops which do not possess an upper bound on the number of iterations. This can lead to resource (gas) exhaustion if there are too many iterations through the loops. An upper bound where appropriate should be applied to avoid these potential concerns.

Resolution:

Loops have been bounded to sensible limits in `68b4c00ff32c7e805967902f229a78d893fc65c3`.

`_getStableCoinInfo` can be clarified

The `_getStableCoinInfo` function comments could/should be clarified, as the error message does not appear to indicate the actual error that should be thrown, rather referencing a downstream event.



Resolution:

Clarification was added in **43b303c3afd78e7eaba11339d4331d74ef1e3426**.

Commented out code should be removed

There are a number of instances within the codebase (such as within fee generation for the reward pool) in which commented out code exists. These instances should be removed.

Resolution:

Commented code was removed in **7342b57fb6a019a6d5986adc58701fc27b151502**.



Toolset Warnings

Unique to the Wolves Finance Protocol

Overview

In addition to our manual review, our process involves utilizing static analysis and formal methods in order to perform additional verification of the presence of security vulnerabilities (or lack thereof). An additional part of this review phase consists of reviewing any automated unit testing frameworks that exist.

The following sections detail warnings generated by the automated tools and confirmation of false positives where applicable.

Compilation Warnings

No compilation warnings were encountered during the course of our audit.

Test Coverage

The contracts possess a number of functional unit tests encompassing various stages of the application lifecycle.

Static Analysis Coverage

The contract repository underwent heavy scrutiny with multiple static analysis agents, including:

- [Securify](#)
- [MAIAN](#)
- [Mythril](#)
- [Oyente](#)
- [Slither](#)



Directory Structure

At time of review, the directory structure of the Wolves Finance smart contracts repository appeared as it does below. Our review, at request of Wolves Finance, covers the Solidity code (*.sol) as of commit hash **ee59f8d7964c3bf943a51f708a064835af2097eb**

```
.
├── booster
│   └── Booster.sol
├── cfolio
│   ├── CFolioItemHandlerLP.sol
│   ├── SFTEvaluator.sol
│   └── interfaces
│       ├── ICFolioItemHandler.sol
│       └── ISFTEvaluator.sol
├── contracts
├── crowdsale
│   ├── Crowdsale.sol
│   └── WOWSSftMinter.sol
├── investment
│   ├── CFolioFarm.sol
│   ├── Controller.sol
│   ├── RewardHandler.sol
│   ├── UniV2StakeFarm.sol
│   └── interfaces
│       ├── ICFolioFarm.sol
│       ├── IController.sol
│       ├── IFarm.sol
│       ├── IRewardHandler.sol
│       └── IStakeFarm.sol
```



```
|   └─ ITxWorker.sol
|── networks.js
|── proxy
|   └─ UpgradeProxy.sol
|── token
|   └─ TradeFloor.sol
|   └─ WOWSCryptofolio.sol
|   └─ WOWSErc1155.sol
|   └─ WOWSErc20.sol
|   └─ WOWSMinterPauser.sol
|   └─ interfaces
|       └─ ICFoliolItemCallback.sol
|       └─ IERC1155BurnMintable.sol
|       └─ IERC20WowsMintable.sol
|       └─ IWOWSCryptofolio.sol
|       └─ IWOWSERC1155.sol
└─ utils
    ├── AddressBook.sol
    ├── AddressRegistry.sol
    ├── ERC20Recovery.sol
    ├── TokenIds.sol
    └─ interfaces
        └─ IAddressRegistry.sol
```

12 directories, 34 files